# Creating Decentralised Applications *with* Blockchain Technology *using* Open Source Tools

By virtue of being a distributed ledger in a P2P network, the blockchain is relatively more secure than data that is on a central server. It is made more secure by the use of powerful cryptography and algorithms. So decentralised applications based on blockchain technology provide excellent security for various types of data.

Blockchain technology is being widely used in the domain of cryptocurrency. A number of digital cryptocurrencies have gained prominence and are being shared throughout the world despite much criticism and controversies. These include Bitcoin, Ethereum, LiteCoin, PeerCoin, GridCoin, PrimeCoin, Ripple, Nxt, DogeCoin, etc.

These blockchain based cryptocurrencies do not have any intermediate bank or payment gateway to record the log of the transactions. That is the main reason why many countries are not recognising cryptocurrencies as valid money transactions. In spite of this, these blockchain based cryptocurrencies are rather popular and are used because of their many security features.

The blockchain network has a block of records, in which every record is associated with dynamic cryptography so that all the transactions can be encrypted without any probability of sniffing or hacking attempts.

## Case studies

In the current scenario, blockchain technology is more focused towards cryptocurrencies in which a distributed ledger is maintained for the transactions. The distributed ledger refers to a replicated and synchronised digital asset shared by users at multiple locations and on various devices so that third party manipulation isn't possible. For example, if a bank uses the distributed ledger with blockchain technology, it